

Project code

```
pitching <- read.csv("pitching_2015_2024.csv")
```

```
#selecting only pitchers with more than 50 IP because they will most likely have the most consistent data  
pitching_selected <- pitching[ pitching$IP > 50, c("WHIP", "ERA", "HR.9", "Age", "K.9", "Team", "Season")]
```

```
knitr::opts_chunk$set(echo = FALSE)  
model1 <- lm(ERA ~ WHIP, data = pitching_selected)  
summary(model1)
```

```
##  
## Call:  
## lm(formula = ERA ~ WHIP, data = pitching_selected)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.64326 -0.45208  0.00269  0.44100  2.70695   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -2.00352    0.08047  -24.90  <2e-16 ***  
## WHIP         4.71282    0.06258   75.31  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.7074 on 3139 degrees of freedom  
## Multiple R-squared:  0.6437, Adjusted R-squared:  0.6436   
## F-statistic: 5671 on 1 and 3139 DF,  p-value: < 2.2e-16
```

```
model2 <- lm(ERA ~ WHIP + HR.9, data = pitching_selected)  
summary(model2)
```

```
##  
## Call:  
## lm(formula = ERA ~ WHIP + HR.9, data = pitching_selected)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.01805 -0.34588 -0.00269  0.33146  2.46692   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -2.05389    0.06273  -32.74  <2e-16 ***  
## WHIP         3.82546    0.05260   72.72  <2e-16 ***  
## HR.9         0.00000    0.00000    0.00  1.000000
```

```

## HR.9          1.03597    0.02300    45.05    <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5514 on 3138 degrees of freedom
## Multiple R-squared:  0.7836, Adjusted R-squared:  0.7835
## F-statistic: 5683 on 2 and 3138 DF,  p-value: < 2.2e-16

## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP
## Model 2: ERA ~ WHIP + HR.9
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    3139 1570.91
## 2    3138  953.97   1     616.95 2029.4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Call:
## lm(formula = ERA ~ WHIP + HR.9 + K.9, data = pitching_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.94523 -0.34570 -0.00743  0.32821  2.51431
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.642426   0.097139  -16.908 < 2e-16 ***
## WHIP         3.709218   0.056419   65.745 < 2e-16 ***
## HR.9         1.028750   0.022926   44.872 < 2e-16 ***
## K.9        -0.029411   0.005319   -5.529 3.48e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5488 on 3137 degrees of freedom
## Multiple R-squared:  0.7857, Adjusted R-squared:  0.7855
## F-statistic: 3834 on 3 and 3137 DF,  p-value: < 2.2e-16

## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9
## Model 2: ERA ~ WHIP + HR.9 + K.9
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    3138  953.97
## 2    3137  944.76   1     9.2074 30.573 3.48e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Call:
## lm(formula = ERA ~ WHIP + HR.9 + K.9 + Age, data = pitching_selected)
##

```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.94239 -0.34125 -0.00674  0.32840  2.52122
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.412398   0.127994 -11.035 < 2e-16 ***
## WHIP         3.695075   0.056593  65.293 < 2e-16 ***
## HR.9         1.032245   0.022937  45.003 < 2e-16 ***
## K.9         -0.030077   0.005319  -5.655 1.7e-08 ***
## Age         -0.007351   0.002667  -2.756 0.00589 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5482 on 3136 degrees of freedom
## Multiple R-squared:  0.7862, Adjusted R-squared:  0.786
## F-statistic: 2884 on 4 and 3136 DF, p-value: < 2.2e-16

## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9 + K.9
## Model 2: ERA ~ WHIP + HR.9 + K.9 + Age
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     3137 944.76
## 2     3136 942.48  1     2.2827 7.5956 0.005885 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##      WHIP      HR.9      K.9      Age
## 1.361824 1.170459 1.213753 1.009367

##
## Call:
## lm(formula = ERA ~ WHIP + HR.9 + K.9 + Age + Team, data = pitching_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.90244 -0.34040 -0.01323  0.33266  2.51326
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.416307   0.140039 -10.114 < 2e-16 ***
## WHIP         3.663800   0.058349  62.791 < 2e-16 ***
## HR.9         1.032689   0.023161  44.587 < 2e-16 ***
## K.9         -0.026143   0.005422  -4.822 1.49e-06 ***
## Age         -0.006634   0.002759  -2.405 0.01624 *
## TeamARI     -0.026835   0.063439  -0.423 0.67232
## TeamATL     -0.095313   0.065257  -1.461 0.14423
## TeamBAL      0.010242   0.063054   0.162 0.87098
## TeamBOS     -0.026569   0.061419  -0.433 0.66534
## TeamCHC     -0.166509   0.065074  -2.559 0.01055 *
## TeamCHW     -0.022786   0.065509  -0.348 0.72799
## TeamCIN     -0.066977   0.064111  -1.045 0.29624
```

```

## TeamCLE      -0.059861    0.060661   -0.987    0.32381
## TeamCOL       0.181370    0.062464    2.904    0.00372 **
## TeamDET       0.090252    0.061193    1.475    0.14035
## TeamHOU      -0.045996    0.061565   -0.747    0.45505
## TeamKCR       0.020813    0.064026    0.325    0.74515
## TeamLAA      -0.043743    0.061960   -0.706    0.48025
## TeamLAD      -0.085015    0.064710   -1.314    0.18902
## TeamMIA       0.041890    0.062576    0.669    0.50327
## TeamMIL      -0.134823    0.061707   -2.185    0.02897 *
## TeamMIN       0.075015    0.063853    1.175    0.24016
## TeamNYM       0.023376    0.064483    0.363    0.71699
## TeamNYY      -0.067943    0.063982   -1.062    0.28836
## TeamOAK       0.150337    0.063936    2.351    0.01877 *
## TeamPHI      -0.014946    0.064183   -0.233    0.81589
## TeamPIT      -0.064058    0.065213   -0.982    0.32603
## TeamSDP       0.014664    0.063768    0.230    0.81814
## TeamSEA       0.023355    0.066711    0.350    0.72629
## TeamSFG       0.015750    0.061232    0.257    0.79702
## TeamSTL      -0.068370    0.065032   -1.051    0.29319
## TeamTBR       0.014778    0.066569    0.222    0.82433
## TeamTEX       0.058152    0.066173    0.879    0.37959
## TeamTOR      -0.053241    0.068515   -0.777    0.43718
## TeamWSN      -0.104343    0.065953   -1.582    0.11373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5463 on 3106 degrees of freedom
## Multiple R-squared:  0.7898, Adjusted R-squared:  0.7875
## F-statistic: 343.2 on 34 and 3106 DF, p-value: < 2.2e-16

## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9 + K.9 + Age
## Model 2: ERA ~ WHIP + HR.9 + K.9 + Age + Team
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1    3136 942.48
## 2    3106 926.83 30      15.65 1.7483 0.007169 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##      WHIP      HR.9      K.9      Age
## 1.361824 1.170459 1.213753 1.009367

##
## Call:
## lm(formula = ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL +
##     OAK, data = pitching_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.93253 -0.34340 -0.00707  0.33689  2.52739
##
## Coefficients:

```

```

##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.421381  0.127768 -11.125 < 2e-16 ***
## WHIP        3.671732  0.056677  64.784 < 2e-16 ***
## HR.9        1.032040  0.022844  45.177 < 2e-16 ***
## K.9        -0.028266  0.005317  -5.316 1.13e-07 ***
## Age        -0.006638  0.002661  -2.495 0.01266 *
## COL        0.192340  0.056947   3.377 0.00074 ***
## CHC        -0.151792  0.059887  -2.535 0.01131 *
## MIL        -0.119609  0.055899  -2.140 0.03245 *
## OAK        0.162808  0.058330   2.791 0.00528 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5459 on 3132 degrees of freedom
## Multiple R-squared:  0.7883, Adjusted R-squared:  0.7878
## F-statistic: 1458 on 8 and 3132 DF,  p-value: < 2.2e-16

## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9 + K.9 + Age
## Model 2: ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL + OAK
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1    3136 942.48
## 2    3132 933.28  4     9.1912 7.7112 3.519e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## OLS estimation, Dep. Var.: ERA
## Observations: 3,141
## Fixed-effects: Season: 10
## Standard-errors: IID
##           Estimate Std. Error  t value    Pr(>|t|)
## WHIP    3.677402    0.056808 64.73347 < 2.2e-16 ***
## HR.9    1.025568    0.023443 43.74766 < 2.2e-16 ***
## K.9   -0.031559    0.005442 -5.79922 7.3279e-09 ***
## Age   -0.007548    0.002664 -2.83307 4.6400e-03 **
## COL    0.190481    0.056839  3.35123 8.1410e-04 ***
## CHC   -0.150833    0.059784 -2.52298 1.1686e-02 *
## MIL   -0.120037    0.055776 -2.15213 3.1463e-02 *
## OAK    0.160636    0.058244  2.75797 5.8500e-03 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.543007      Adj. R2: 0.788809
##                               Within R2: 0.785632

##
## Hausman Test
##
## data: ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL + OAK
## chisq = 24.297, df = 8, p-value = 0.002043
## alternative hypothesis: one model is inconsistent

## OLS estimation, Dep. Var.: ERA

```

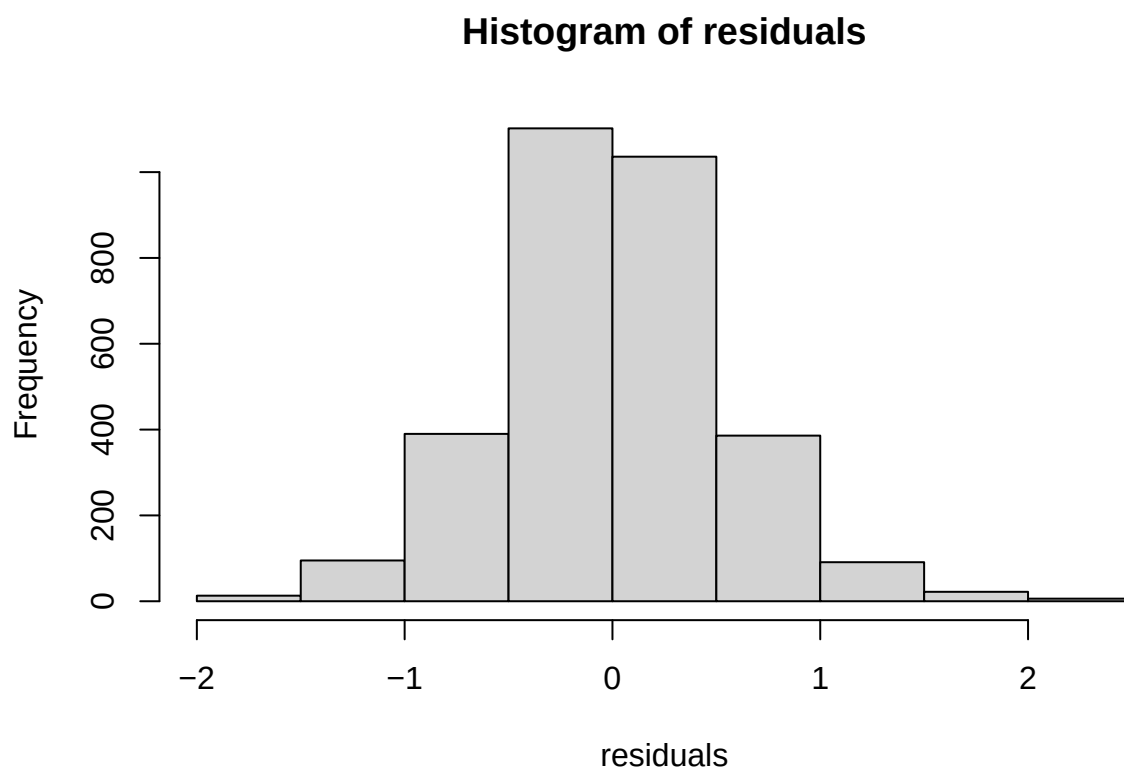
```

## Observations: 3,141
## Fixed-effects: Season: 10
## Standard-errors: IID
##      Estimate Std. Error  t value   Pr(>|t|)
## WHIP  3.677402   0.056808 64.73347 < 2.2e-16 ***
## HR.9   1.025568   0.023443 43.74766 < 2.2e-16 ***
## K.9   -0.031559   0.005442 -5.79922 7.3279e-09 ***
## Age   -0.007548   0.002664 -2.83307 4.6400e-03 **
## COL    0.190481   0.056839  3.35123 8.1410e-04 ***
## CHC   -0.150833   0.059784 -2.52298 1.1686e-02 *
## MIL   -0.120037   0.055776 -2.15213 3.1463e-02 *
## OAK    0.160636   0.058244  2.75797 5.8500e-03 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.543007      Adj. R2: 0.788809
##                      Within R2: 0.785632

## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9 + K.9 + Age
## Model 2: ERA ~ WHIP + HR.9 + K.9 + Age + BB.9 + AVG
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1    3136 942.48
## 2    3134 879.91  2    62.569 111.43 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

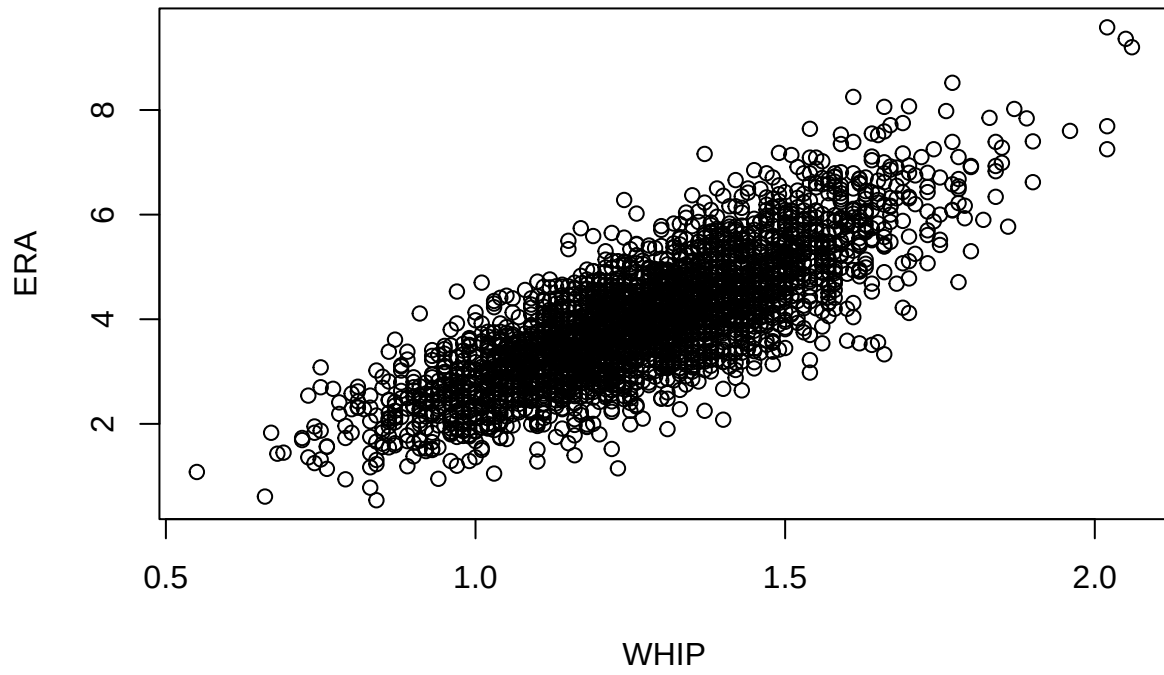
##      WHIP      HR.9      K.9      Age      BB.9      AVG
## 144.342276  1.268922  1.770340  1.012805  43.881251 103.329146

```

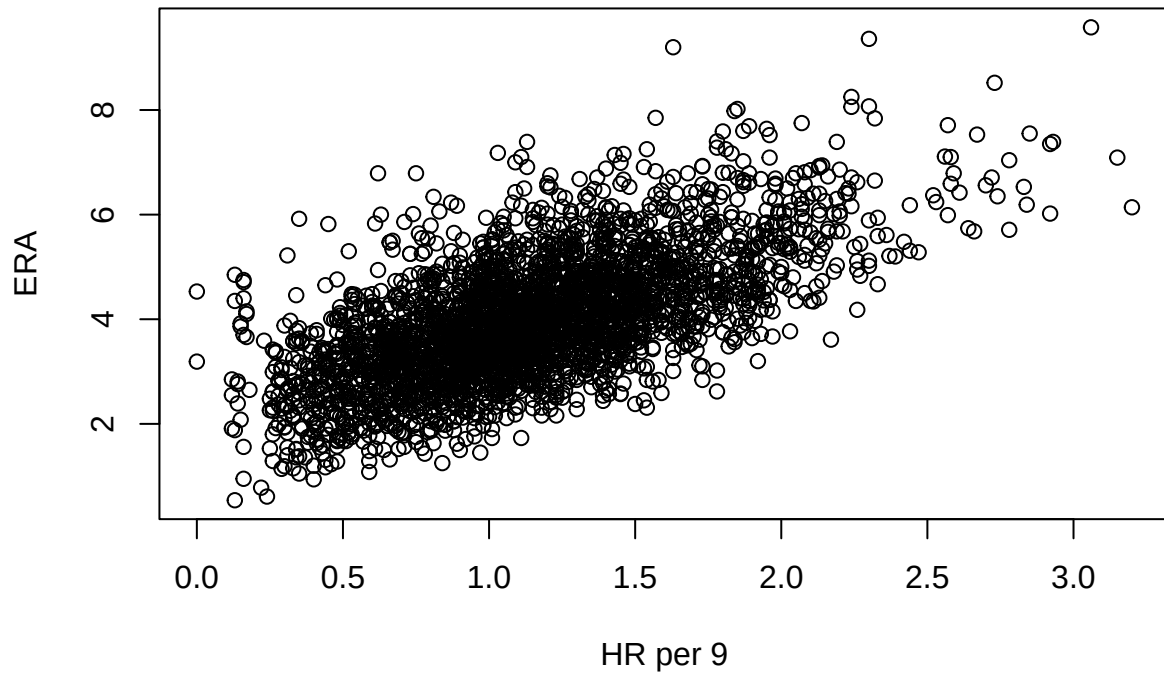


```
##  
## Anderson-Darling normality test  
##  
## data: residuals  
## A = 3.8137, p-value = 1.665e-09
```

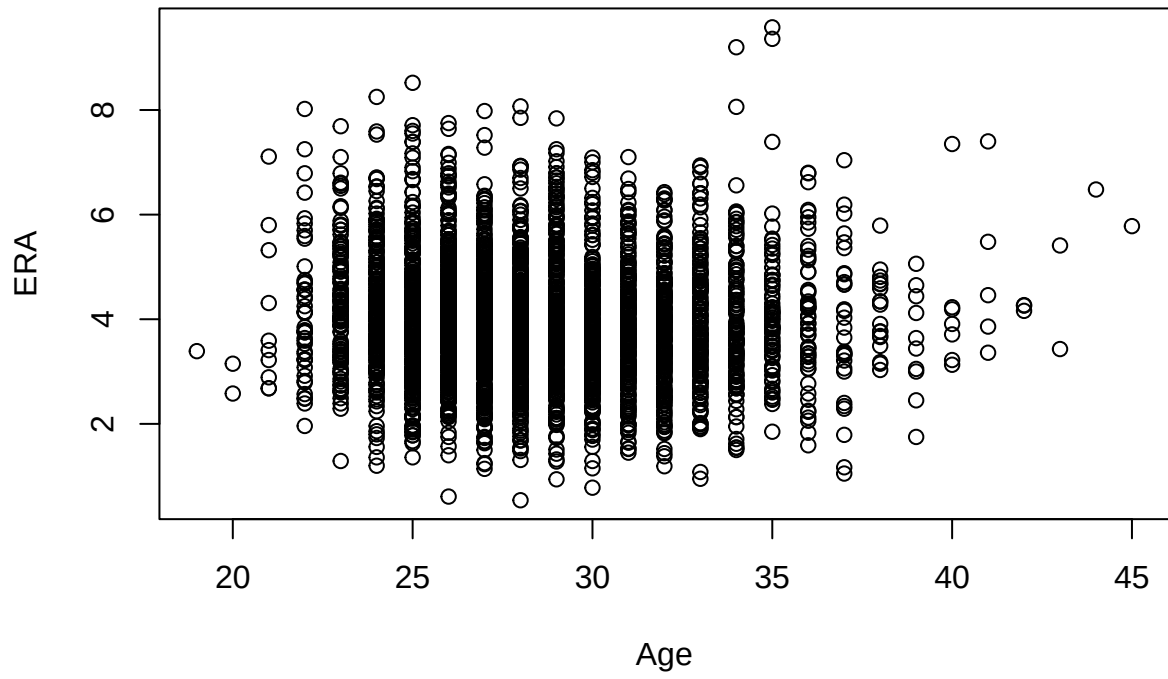
Relationship Between WHIP and ERA



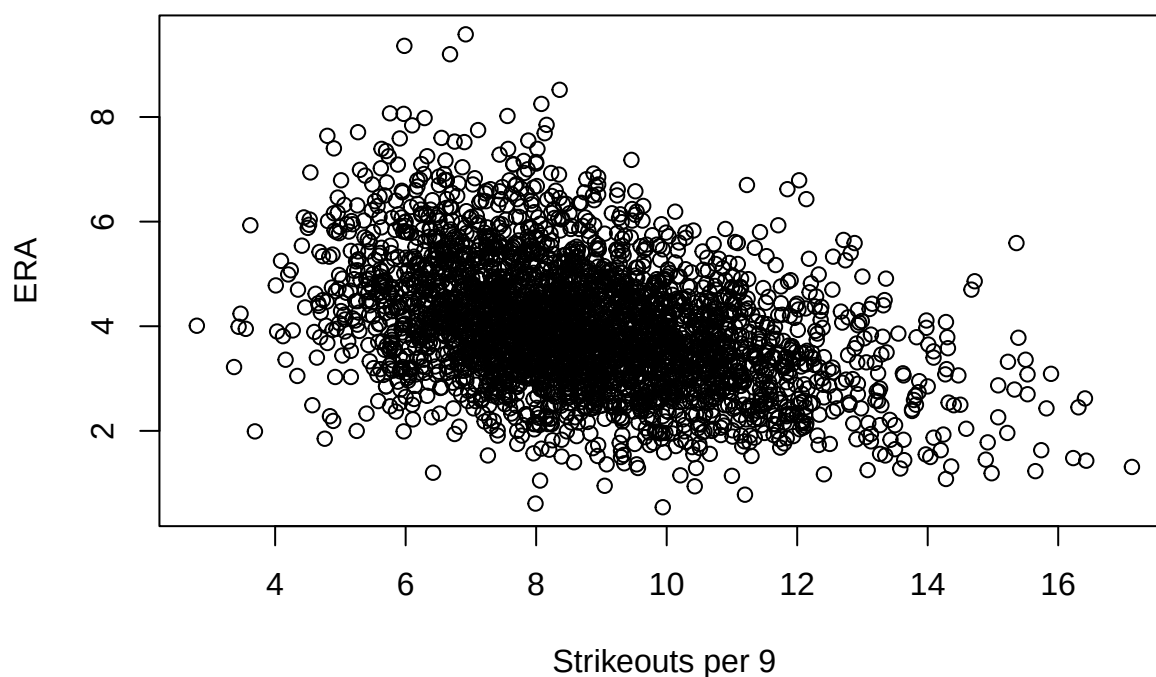
Relationship Between HRs per 9 and ERA



Relationship Between Age and ERA



Relationship Between Strikeouts per 9 and ERA



```
## [1] 0.001758078
```

```
## OLS estimation, Dep. Var.: ERA
```

```
## Observations: 3,141
```

```
## Fixed-effects: Season: 10
```

```
## Standard-errors: IID
```

	Estimate	Std. Error	t value	Pr(> t)
## WHIP	3.691340	0.056659	65.15065	< 2.2e-16 ***
## HR.9	1.022454	0.023443	43.61381	< 2.2e-16 ***
## K.9	-0.030662	0.005439	-5.63755	1.8784e-08 ***
## COL	0.194912	0.056881	3.42665	6.1898e-04 ***
## CHC	-0.159742	0.059768	-2.67268	7.5638e-03 **
## MIL	-0.120015	0.055838	-2.14933	3.1685e-02 *
## OAK	0.162749	0.058305	2.79133	5.2811e-03 **

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## RMSE: 0.543705      Adj. R2: 0.788334
```

```
##                    Within R2: 0.785081
```

```
## [1] 0.02015268
```

```
## Warning: There was 1 warning in 'summarise()'.
```

```
## i In argument: 'across(...)'.
```

```
## Caused by warning:
```

```
## ! The '...' argument of 'across()' is deprecated as of dplyr 1.1.0.
```

```

## Supply arguments directly to '.fns' through an anonymous function instead.
##
## # Previously
## across(a:b, mean, na.rm = TRUE)
##
## # Now
## across(a:b, \(x) mean(x, na.rm = TRUE))

## WHIP_mean WHIP_min WHIP_max WHIP_median WHIP_sd ERA_mean ERA_min ERA_max
## 1 1.269968 0.55 2.06 1.27 0.2017359 3.981621 0.54 9.58
## ERA_median ERA_sd HR.9_mean HR.9_min HR.9_max HR.9_median HR.9_sd
## 1 3.89 1.184993 1.136434 0 3.2 1.1 0.4614473
## Age_mean Age_min Age_max Age_median Age_sd K.9_mean K.9_min K.9_max
## 1 28.60331 19 45 28 3.685214 8.692305 2.8 17.13
## K.9_median K.9_sd BB.9_mean BB.9_min BB.9_max BB.9_median BB.9_sd AVG_mean
## 1 8.52 2.026344 3.116129 0.6 8.25 3.03 1.00435 0.2395664
## AVG_min AVG_max AVG_median AVG_sd COL_mean COL_min COL_max COL_median
## 1 0.111 0.349 0.241 0.03445017 0.03088188 0 1 0
## COL_sd CHC_mean CHC_min CHC_max CHC_median CHC_sd MIL_mean MIL_min
## 1 0.1730252 0.02737982 0 1 0 0.1632135 0.03151862 0
## MIL_max MIL_median MIL_sd OAK_mean OAK_min OAK_max OAK_median OAK_sd
## 1 1 0 0.1747424 0.02897167 0 1 0 0.1677536
## ERA_resid2_mean ERA_resid2_min ERA_resid2_max ERA_resid2_median ERA_resid2_sd
## 1 1.171841e-16 -3.446135 5.444855 -0.08177966 1.172992

```